

Studio based Project - AI, Cloud, and Big Data

Dataset Documentation

Project title: _____

Project group details:

Sr. No.	Enrolment Number(s)	Student Name(s)
1		
2		
3		
4		

Submission Date: _____

Few guidelines and pointers about how to and what to fill in the information

1. General dataset information

- a. Name of dataset - A clear, concise title.
- b. Source or origin - Where did the data come from? (e.g., Kaggle, UCI Repository, API, web scraping, survey, internal organization).
- c. Creator(s) and ownership - Who collected or created the data?
- d. Publication date and access date - When was it created and when did the student acquire it?
- e. Licensing and terms of use - Are there restrictions, copyrights, or requirements for attribution?
- f. Reference/citation - A link or citation to the original source. (APA format)

2. Contextual information

- a. Description - A brief summary of what the data represents and the problem it aims to solve.
- b. Keywords - Descriptive terms for categorization (e.g., healthcare, finance, NLP, computer vision).
- c. Time period (temporal coverage) - What time range does the data represent (e.g., sales from 2020-2025)?
- d. Geographic coverage - Is the data limited to a specific region like country, states, cities?

3. Technical structure and contents

- a. File formats and structure - Formats used (CSV, JSON, SQL, JPG, TXT, MP4) and file size.
- b. Type of data - Structured (rows/columns) or Unstructured (text, images, audio).
- c. Data dictionary (For structured data)

- Column names, data types (numerical, categorical, date), units of measurement, and definitions of each variable.
 - Codes or abbreviations used.
- d. Sample description (for unstructured data) - Details on image resolution, audio duration, or text length.
- e. Total Count (Volume) - Total number of rows, records, or files.

4. Data quality and preprocessing

- a. Missing values - Description of missing data, including codes (e.g., NULL, NA, 999).
- b. Known Issues (limitations) - Any known bugs, biases, or gaps in the data.
- c. Data cleaning and preprocessing steps - What steps were taken to make the data usable (e.g., removing duplicates, handling missing values, normalization)?
- d. Annotation or labeling (For ML projects) - If it is a labeled dataset, describe how the labels were generated.

Checklist

1. Basic Dataset Information	5
2. Data Characteristics	5
3. Big Data Perspective (5 V's).....	5
4. Data Quality Assessment.....	6
5. Data Preprocessing Details	6
6. Exploratory Data Analysis (EDA).....	6
7. Tools and Technologies Used.....	6
8. Ethical and Legal Considerations	7
9. Data Storage and Management.....	7
10. Problem Definition Based on Dataset.....	7
11. Data Splitting Strategy	7

Note: From the above 11 points, prepare a documentation for first 9 points. Point numbers 10 and 11 can be filled in later stage after discussion with Prof. Jaideep sir.

1. Basic dataset information

Dataset name: _____

Domain / application area: _____

Type of data (structured / semi-structured / unstructured): _____

Source of dataset (Kaggle, UCI, API, Survey, etc.): _____

Dataset link / reference: _____

License / usage rights: _____

2. Data characteristics

For Structured Data:

- Number of records (rows): _____
- Number of features (columns): _____
- Attribute names and description: *Add a table here if required*
- Data types (numeric, categorical, etc.): _____
- Target variable (if applicable): _____

For Unstructured Data:

- Total number of files: _____
- File format (JPEG, TXT, MP4, etc.): _____
- Total dataset size (MB/GB): _____
- Annotation availability (Labeled/Unlabeled): _____

3. Big Data perspective (5 Vs)

Volume: _____

Velocity: _____

Variety: _____

Veracity: _____

Value: _____

4. Data quality assessment

Missing values analysis (pure and impure data): _____

Duplicate records: _____

Outliers: _____

Noise / inconsistencies: _____

Data imbalance (if applicable): _____

Bias considerations: _____

5. Data preprocessing details

Data cleaning techniques used: _____

Handling missing values: _____

Encoding techniques: _____

Feature scaling / normalization: _____

Transformation steps: _____

6. Exploratory Data Analysis (EDA)

Summary statistics: _____

Visualization techniques used: _____

Correlation analysis: _____

Key observations: _____

7. Tools and technologies used

Programming language(s): _____

Libraries / frameworks: _____

Visualization tools: _____

Big data tools (if any): _____

Any other tool / technology: _____

8. Ethical and legal considerations

Personal data involved: _____

Consent obtained: _____

Anonymization techniques used: _____

Compliance with data protection norms: _____

Sensitive Attributes if any (e.g., race, gender, religion): _____

9. Data storage and management

Storage type (Local / Cloud / HDFS / Database): _____

Storage platform used (AWS, Azure, Google Cloud, Hadoop HDFS, etc.): _____

Data format stored (CSV, Parquet, JSON, ORC, etc.): _____

Data organization structure (Folders, Partitions, Buckets, etc.): _____

Backup and recovery strategy: _____

Security measures (Encryption, Access Control, IAM, etc.): _____

Scalability considerations: _____

Ingestion method (How is the data moved from the source to the storage? (e.g., Manual upload, Python scripts, AWS Glue, Azure Data Factory) _____

10. Problem definition based on dataset

Type of problem (Classification/Regression/Clustering/etc.): _____

Problem statement: _____

Justification for dataset selection: _____

11. Data splitting strategy

Train/Test split ratio: _____

Validation strategy (K-Fold, etc.): _____

Sampling technique used: _____

In future, you will address the points like - Data pipeline architecture, API integration details, Real-time/batch/stream processing, Version Control, data versioning, code repository, Success metrics, technical architecture etc.